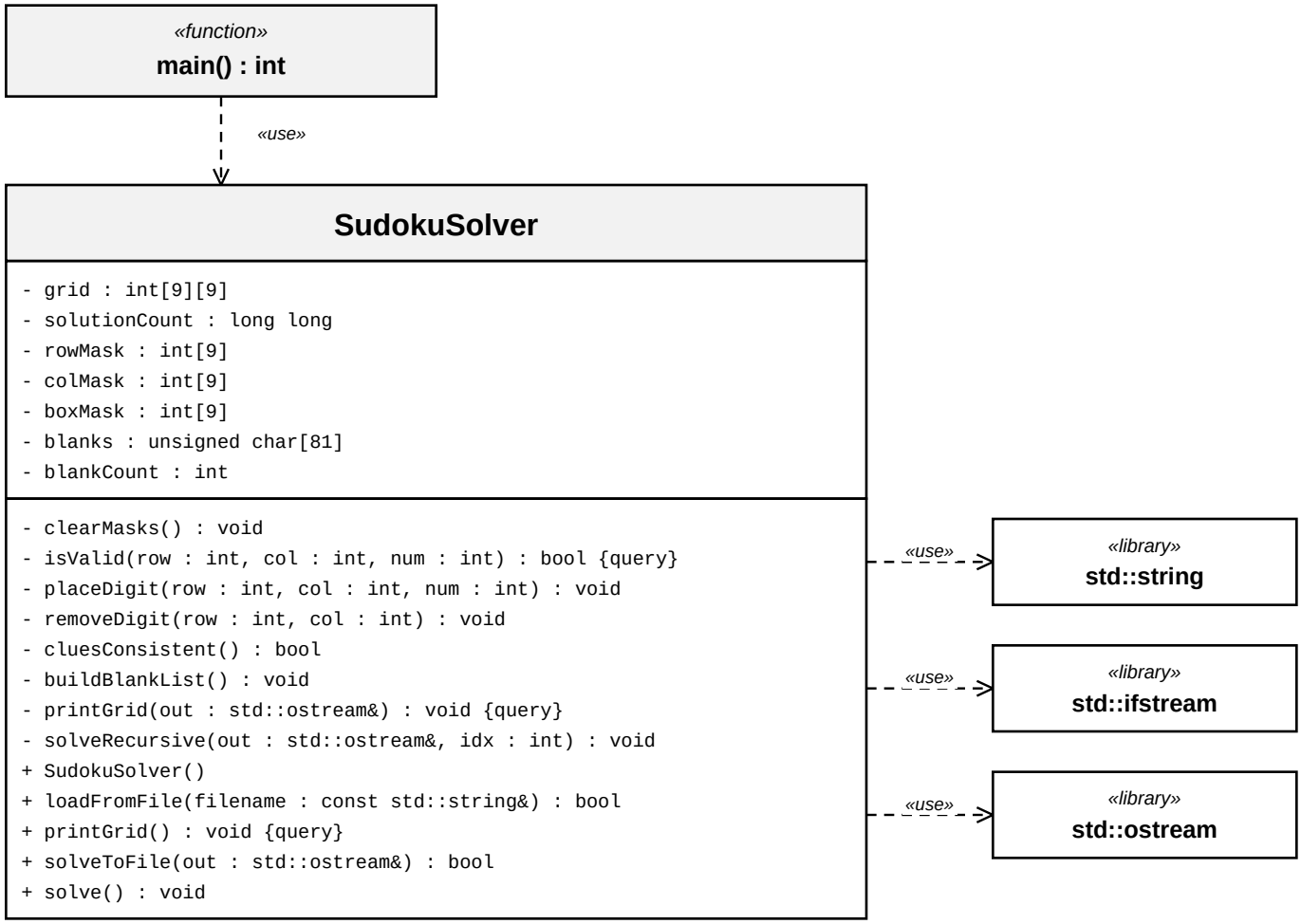
Assignment 6 - UML Class Diagrams

EECS 348 Software Engineering I (Fall 2026) | Josselyn Bui | Program: Sudoku Solver (Assignment6.cpp) | November 12, 2026
Diagrams created with GenAI assistance (Claude, Anthropic) and reviewed by the author. Visibility: + public, - private. {query} marks a const member function.

Class diagram: SudokuSolver (the only class in the program)



main() is a free function, not a class, so it has no class diagram of its own. It is drawn only to show who uses SudokuSolver. The standard library types are shown as library classes to document the relationships.

Attributes of SudokuSolver

Visibility	Name	Type	Purpose
private (-)	`grid`	`int[9][9]`	The Sudoku board; 0 means the cell is empty.
private (-)	`solutionCount`	`long long`	Number of complete solutions found for the current puzzle.
private (-)	`rowMask`	`int[9]`	Bit d is set when digit d is already used in that row.
private (-)	`colMask`	`int[9]`	Bit d is set when digit d is already used in that column.
private (-)	`boxMask`	`int[9]`	Bit d is set when digit d is already used in that 3x3 box.
private (-)	`blanks`	`unsigned char[81]`	Position (0 to 80) of each blank cell, in row-major order.
private (-)	`blankCount`	`int`	How many entries of blanks are in use.

Methods of SudokuSolver

Visibility	Name	Parameter list	Return type	Purpose
private (-)	`clearMasks`	`()`	`void`	Resets the row, column, and box masks to no digits used.
private (-)	`isValid`	`(row : int, col : int, num : int)`	`bool {query}`	True when num is not blocked in the cell's row, column, or box.
private (-)	`placeDigit`	`(row : int, col : int, num : int)`	`void`	Writes a digit into the board and updates the three masks.
private (-)	`removeDigit`	`(row : int, col : int)`	`void`	Clears a cell and removes its digit from the three masks (backtrack).
private (-)	`cluesConsistent`	`()`	`bool`	Builds the masks from the clues; false if two clues conflict.
private (-)	`buildBlankList`	`()`	`void`	Records the position of every blank cell once, in row-major order.
private (-)	`printGrid`	`(out : std::ostream&)`	`void {query}`	Writes the board to the given stream in the input-file format.
private (-)	`solveRecursive`	`(out : std::ostream&, idx : int)`	`void`	Backtracking search over the blank list; counts and writes every solution.
public (+)	`SudokuSolver`	`()`	`-`	Constructor. Creates an empty board with zero solutions and no digits used.
public (+)	`loadFromFile`	`(filename : const std::string&)`	`bool`	Reads exactly 81 valid tokens; false for a missing or malformed file.
public (+)	`printGrid`	`()`	`void {query}`	Prints the original puzzle to the console.
public (+)	`solveToFile`	`(out : std::ostream&)`	`bool`	Finds all solutions and writes them with the count or no-solution line.
public (+)	`solve`	`()`	`void`	Compatibility method; solves and prints to the console.

Relationships

From	To	Relationship	Notes
main()	SudokuSolver	Dependency «use»	main() creates one SudokuSolver per puzzle and calls loadFromFile, printGrid, and solveToFile.
SudokuSolver	std::string	Dependency «use»	Parameter type of loadFromFile.
SudokuSolver	std::ifstream	Dependency «use»	Local object in loadFromFile that reads the puzzle file.
SudokuSolver	std::ostream	Dependency «use»	Parameter type of printGrid, solveRecursive, and solveToFile; main() passes the shared solutions.txt stream or std::cout.
Other user classes	-	None	SudokuSolver is the only user-defined class, so there is no inheritance, association, aggregation, or composition.